Martin–Solovay Construction

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Theorem (Martin–Solovay)**.** *Martin's Axiom (or* MA *for short) implies that* $2^\kappa = \mathfrak{c}$ *for all infinite cardinals* $\kappa < \mathfrak{c}$*, and consequently that* $\mathfrak{c}$ *is regular.*

Proof. Assuming MA, we first show that $2^\kappa \leq \mathfrak{c}$ for every infinite cardinal $\kappa < \mathfrak{c}$.

Fix an infinite cardinal $\kappa < \mathfrak{c}$. Since there exists an almost disjoint family of size $\mathfrak{c}$, we can select an almost disjoint family $\mathcal{A} = \{A_\alpha\}_{\alpha<\kappa}$ of size κ.

For any $X \subseteq \kappa$, our goal is to construct a set $B_X \subseteq \omega$ such that for all $\alpha < \kappa$,

$$\alpha \in X \iff B_X \cap A_\alpha \text{ is infinite.} \tag{$*$}$$

If every subset $X \subseteq \kappa$ is uniquely determined by such a $B_X \subseteq \omega$ via

$$X = \{\alpha < \kappa \mid B_X \cap A_\alpha \text{ is infinite}\},$$

it yields a surjection from $\mathcal{P}(\omega)$ onto $\mathcal{P}(\kappa)$. Thus, $2^\kappa \leq 2^{\aleph_0} = \mathfrak{c}$.

To construct B_X for a fixed $X \subseteq \kappa$ and almost disjoint family $\mathcal{A}$, define the partial order $(P^X_{\mathcal{A}}, \leq^X_{\mathcal{A}})$ as follows:

- Conditions are functions p satisfying:
  - $p \subseteq \omega \times \{0, 1\}$,
  - $\mathrm{dom}(p) \cap A_\alpha$ is finite for every $\alpha \in X$,
  - $p_+ := \{n \in \mathrm{dom}(p) \mid p(n) = 1\}$ is finite.
- The ordering is reverse inclusion ($p \leq^X_{\mathcal{A}} q \iff q \subseteq p$).

Given a family $\mathcal{D}$ of dense sets, every $\mathcal{D}$-generic filter $G \subseteq P^X_{\mathcal{A}}$ yields a corresponding subset $B_X := \bigcup\{p_+ \mid p \in G\}$ of ω. To ensure B_X satisfies ($*$), we construct $\mathcal{D}$ from the following dense sets for each $\alpha < \kappa$:

- For $\alpha \in X$ and $n \in \omega$, let $D'_{\alpha,n} := \{p \in P^X_{\mathcal{A}} \mid p_+ \cap A_\alpha \text{ is of size at least } n\}$. Meeting these sets forces $B_X \cap A_\alpha$ to be infinite.
- For $\alpha \notin X$, let $D''_\alpha := \{p \in P^X_{\mathcal{A}} \mid A_\alpha \subseteq \mathrm{dom}(p)\}$. Meeting these sets forces $B_X \cap A_\alpha$ to be finite.

Since $|\mathcal{D}| = \kappa < \mathfrak{c}$, MA will guarantee the existence of such a $\mathcal{D}$-generic filter G, provided we show that our poset satisfies the countable chain condition and that $\mathcal{D}$ is indeed a family of dense sets.

To verify the ccc, note that if $p_+ = q_+$, then p and q are compatible (since $p \cup q$ forms a valid condition). Thus, any antichain in our poset can contain at most one condition for each choice of p_+. Since p_+ is a finite subset of ω, there are only countably many such choices, so $(P^X_{\mathcal{A}}, \leq^X_{\mathcal{A}})$ is ccc. To verify density:

- For $D'_{\alpha,n}$ (where $\alpha \in X$ and $n \in \omega$): Given $p \in P^X_{\mathcal{A}}$, since $\mathrm{Dom}(p) \cap A_\alpha$ is finite, $A_\alpha \setminus \mathrm{Dom}(p)$ is infinite. Pick distinct $k_1, \ldots, k_n \in A_\alpha \setminus \mathrm{Dom}(p)$ and extend p to p' by setting $p'(k_1) = \cdots = p'(k_n) = 1$. Since only finitely many values were added, the extension p' is a valid condition in $P^X_{\mathcal{A}}$ belonging to $D'_{\alpha,n}$.
- For D''_α (where $\alpha \notin X$): Given $p \in P^X_{\mathcal{A}}$, extend p to p' by setting $p'(k) = 0$ for all $k \in A_\alpha \setminus \mathrm{Dom}(p)$. Since $p'_+ = p_+$ and $\mathrm{Dom}(p')$ extends A_α by only finitely many elements, p' remains a valid condition in $P^X_{\mathcal{A}}$ with $A_\alpha \subseteq \mathrm{Dom}(p')$, meaning $p' \in D''_\alpha$.

Let $G \subseteq P^X_{\mathcal{A}}$ be the $\mathcal{D}$-generic filter guaranteed by MA. We verify that B_X satisfies (∗):

($\Rightarrow$) Let $\alpha \in X$. For each $n \in \omega$, G intersects $D'_{\alpha,n}$ in some condition p, so $p_+ \cap A_\alpha$ has at least n elements. Since $p_+ \subseteq B_X$, it follows that $B_X \cap A_\alpha$ is of size at least n for all $n \in \omega$, so $B_X \cap A_\alpha$ is infinite.

($\Leftarrow$) Let $\alpha \in \kappa \setminus X$. Since G intersects D''_α, pick $p \in G \cap D''_\alpha$. Then $A_\alpha \subseteq \mathrm{Dom}(p)$, and since all conditions in G are compatible with p, no element of G can set 1 on $A_\alpha \setminus p_+$. Thus $B_X \cap A_\alpha \subseteq p_+$, which is finite.

Finally, $\mathfrak{c}$ is regular, since

$$\mathrm{cof}(\mathfrak{c}) = \mathrm{cof}(2^\kappa) > \kappa \text{ for all } \kappa < \mathfrak{c}.$$

□